CLASS 10 MATHEMATICS • CHAPTER 3

# Pair of Linear Equations in Two Variables

*Exhaustive point-by-point presentation covering graphical and algebraic methods directly from the textbook.*

🎓 Akash Srivastva (www.akashmaths.online)

# Introduction & Real-Life Modeling

## 🎟 Akhila's Fair Visit

- Akhila went to a fair, enjoyed the Giant Wheel and played Hoopla.
- Number of Hoopla games (y) is half the number of Giant Wheel rides (x).
- Giant Wheel ride costs ₹3, Hoopla costs ₹4, total spent ₹20.

## √× Algebraic Equations Formed

$$y = (1/2)x \Rightarrow x - 2y = 0$$
$$3x + 4y = 20$$

Can we find the simultaneous solution? Let's explore graphical and algebraic techniques!

# Section 3.1 Live Check

Q1. How did we translate Akhila playing Hoopla half the number of times she rode the Giant Wheel into an equation?

Q2. What is the general standard form of a linear equation in two variables x and y?

Q3. Can trial and error be efficient for large values, or do we need systematic graphical/algebraic methods?

# Consistent, Inconsistent & Dependent Pairs

## Key Terminology

- **Consistent Pair:** Has at least one solution (intersecting or coincident lines).
- **Inconsistent Pair:** Has no solution (parallel lines).
- **Dependent Pair:** Equivalent equations with infinitely many solutions (always consistent).

## Graphical Behaviors

- **Intersecting lines:** Unique solution (Consistent).
- **Parallel lines:** No solution (Inconsistent).
- **Coincident lines:** Infinitely many solutions (Dependent/Consistent).

# Consistency Concepts Live Check

Q1. What is the difference between a consistent pair and an inconsistent pair of linear equations?

Q2. Are dependent pairs of linear equations also consistent? Why or why not?

Q3. If two lines are parallel when plotted on a graph, what does that indicate about their solutions?

# Comparing Coefficient Ratios

**General Form: $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$**

We compare ratios $a_1/a_2$, $b_1/b_2$, and $c_1/c_2$ to determine graphical & algebraic nature:

1. $(a_1/a_2) \neq (b_1/b_2)$ ⇒ Intersecting lines ⇒ Unique solution (Consistent)
2. $(a_1/a_2) = (b_1/b_2) = (c_1/c_2)$ ⇒ Coincident lines ⇒ Infinitely many solutions (Dependent)
3. $(a_1/a_2) = (b_1/b_2) \neq (c_1/c_2)$ ⇒ Parallel lines ⇒ No solution (Inconsistent)

# Ratio Rules Live Check

Q1. What condition on the ratios $a_1/a_2$ and $b_1/b_2$ implies that the lines intersect at exactly one point?

Q2. If $a_1/a_2 = b_1/b_2 = c_1/c_2$, what can you say about the lines and their solutions?

Q3. When do we get parallel lines and no solutions based on the coefficients?

# Graphical Solutions: Examples 1, 2 & 3

### Example 1: x + 3y = 6 & 2x - 3y = 12

- Intersect at (6, 0). Consistent!
- Solution: x = 6, y = 0.

### Examples 2 & 3

- **Ex 2:** 5x – 8y + 1 = 0 and 3x – (24/5)y + (3/5) = 0 give coincident lines (infinitely many solutions).
- **Ex 3 (Champa's Pants & Skirts):** y = 2x – 2 & y = 4x – 4 intersect at (1, 0) (1 pant, 0 skirts).

# Graphical Examples Live Check

Q1. How do we verify graphically whether a pair of linear equations is consistent?

Q2. What happens when two equations represent the exact same line when plotted?

Q3. How do we check if coordinates like (1, 0) satisfy both algebraic equations in a word problem?

# The Substitution Method

## Step-by-Step Procedure

- **Step 1:** Find the value of one variable (e.g., y) in terms of x from either equation.
- **Step 2:** Substitute this value in the other equation to get an equation in one variable.
- **Step 3:** Solve for that variable, then substitute back to find the other.

## Special Outcomes

- True statement with no variables (e.g., 18 = 18): Infinitely many solutions.
- False statement with no variables (e.g., −4 = 0): Inconsistent pair (no solution).

# Substitution Method Live Check

Q1. Why is the substitution method preferred over graphical methods when solutions have non-integral coordinates?

Q2. What does arriving at a false statement like −4 = 0 tell us during substitution?

Q3. Can you outline the three steps involved in solving by substitution?

# The Elimination Method

### How It Works

- Multiply both equations by suitable non-zero constants to make the coefficients of one variable numerically equal.
- Add or subtract equations to eliminate that variable.
- Solve the resulting single-variable equation.

### Example 8 Highlights

- Incomes 9x & 7x, expenditures 4y & 3y, savings ₹2000.
- Eliminating y gives x = 2000.
- Monthly incomes: ₹18,000 and ₹14,000.

# Elimination Method Live Check

Q1. What is the core idea behind the elimination method for solving linear equations?

Q2. Why do we sometimes need to multiply equations by constants before adding or subtracting them?

Q3. If elimination yields 0 = 9 (a false statement), what conclusion do we draw about the system?

# Worked Exercise Highlights

### Problem 1: Garden Perimeter (Ex 3.1 Q5)

- Length is 4m more than width: $x = y + 4$.
- Half perimeter is 36m: $x + y = 36$.
- Substitute x: $(y + 4) + y = 36 \Rightarrow 2y = 32 \Rightarrow y = 16$m.
- **Dimensions: Length = 20m, Width = 16m.**

### Problem 2: Two-Digit Number (Ex 3.2 Q2(iii))

- Sum of digits is 9: $x + y = 9$.
- 9 times number = 2 times reversed: $9(10x+y) = 2(10y+x)$.
- Simplifies to $8x = y$. Solving gives $x = 1, y = 8$.
- **Number: 18.**

# Exercise Highlights Live Check

Q1. How do we express a two-digit number algebraically using its ten's digit x and unit's digit y?

Q2. In rectangle word problems, how does knowing half the perimeter simplifies our equations?

Q3. When reversing digits of a number $10x + y$, what does the new number become?

# Chapter Recap & Checklist

## Graphical Conditions

- **Intersecting:** $(a_1/a_2) \neq (b_1/b_2)$ → Unique solution (Consistent).
- **Coincident:** $(a_1/a_2) = (b_1/b_2) = (c_1/c_2)$ → Infinitely many solutions (Dependent).
- **Parallel:** $(a_1/a_2) = (b_1/b_2) \neq (c_1/c_2)$ → No solution (Inconsistent).

## Algebraic Methods

- **Substitution Method:** Isolate one variable and substitute.
- **Elimination Method:** Equalize coefficients and add/subtract.

# THANK YOU

*"Equations are more important to me, because politics is for the present, while an equation is something for eternity." — Albert Einstein*

🎓 Akash Srivastva (www.akashmaths.online)